

Poster: GEM-PLI Side Channel in GPON: Inferring Encrypted Streaming Traffic

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Abstract—Gigabit Passive Optical Networks (GPON), an ITU-T standardized technology widely adopted in fiber-to-the-home (FTTH) networks, supports downstream payload encryption to protect subscribers’ traffic. Unfortunately, encryption does not cover all access-layer metadata. This poster presents a preliminary analysis showing that the Payload Length Indicator (PLI), which remains in plaintext in GEM headers, forms a side channel that enables inference of streaming activity. Using a GPON downstream simulator and controlled runs of youtube streaming, our preliminary results show that PLI traces are highly consistent across repeated runs of the same video and suggest that a passive attacker on the same optical splitter as the victim could potentially infer a victim’s streaming activity (e.g., Youtube Video Identification).

I. INTRODUCTION AND MOTIVATION

As fiber internet access has become dominant in Network Infrastructures, GPON—one of the passive optical network (PON) technologies—is adopted by many ISPs to provide fiber-to-the-home (FTTH). Since GPON is popular PON technology, it is important to ensure traffic confidentiality. GPON supports AES-based downstream traffic encryption on GPON Encapsulation Method (GEM) Layer. However, some fields of GEM Header remain plaintext such as the Payload Length Indicator (PLI) field used for multiplexing and the Port-ID field used for mapping subscribers’ Optical Network Terminals/Units(ONTs/ONUs).

Such plaintext fields may appear innocuous. However, recent studies have demonstrated that such metadata can be exploited to infer victim’s activity (e.g., identifying a victim’s web-based video streaming watching activity) [4] [5] [6].

While prior studies primarily focused on upper-layer or cellular traffic analysis [4], [5], We researched whether the plaintext PLI field in GPON GEM headers can be used to leak victim’s privacy on the same optical splitter. We conduct a preliminary experimental analysis using a software-based GPON downstream simulator to evaluate whether the plaintext PLI field could contribute to privacy leakage. Using these PLI traces, we present preliminary evidence that YouTube streaming traces are highly consistent across repeated runs and may be distinguishable in controlled settings.

II. BACKGROUND

GPON Topology and GTC Framing. GPON Networks consist of an Optical Line Terminal (OLT) at the ISP and multiple Optical Network Terminals/Units(ONTs/ONUs) (typically called “modem”) deployed at subscribers. A single

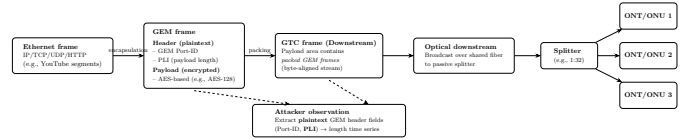


Fig. 1. Encapsulation pipeline in GPON downstream: upper-layer packets (e.g., Ethernet/IP) are encapsulated into GEM frames, packed into GTC frames, and broadcast over the shared optical line.

optical tree can serve up to 32 subscribers (e.g., 1:32 split). In the downstream, the OLT transmits downstream traffic through the splitter. Since the splitter passively distributes the optical signal from the OLT to all ONUs/ONTs on the same splitter, each ONU/ONT can physically receive the downstream transmissions destined for other ONUs/ONTs. At the transmission layer, downstream bytes are carried in GPON Transmission Convergence (GTC) frames sent periodically(per-125 μ s), whose payload area carries multiple GEM frames(described next) [2], [1].

GEM Framing. Upper layer traffic (e.g., Ethernet frames carrying TCP/IP packets) are carried in the GEM Payload Field transmitted via GPON Encapsulation Method (GEM) Frames. Downstream GEM payloads are often AES encrypted for confidentiality (depending on ISP configuration [1]). However, the GEM Port-ID (used to demultiplex services/flows) and the Payload Length Indicator (PLI), which encodes the payload size, are transmitted in plaintext. Thus, an attacker can extract a victim’s encrypted payload lengths from consecutive PLI values (optionally grouped by Port-ID), even without decrypting the payload contents.

Video Identification. HTTP Adaptive Streaming (HAS) is the popular video streaming model – which is generally used by video streaming platforms (e.g., Youtube) –delivers video by letting the client periodically request short media segments at a bitrate chosen by an adaptive bitrate (ABR) algorithm. This request–response process forms characteristic bursty download patterns and bitrate-dependent segment sizes, which can be exploited for traffic inference. [4]

III. THREAT MODEL

We consider an attacker on the same optical splitter as a victim. And we assume an attacker can obtain the victim’s Port-ID.

- **1. Premise** The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can

receive raw GTC frames) connected to splitter and passively records downstream from OLT. the attacker does not need to register their device with OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting.

- **2. Collect GEMs** Then, attacker extracts GTC frames from downstream traffic and parses GEM frames.
- **3. Convert GEM PLI to dataset** Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps.
- **4. Dataset format** Each record includes `t_start_us`, `gem_id`, `port_id`, and `pli`.
- **5. Classification via database matching** The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

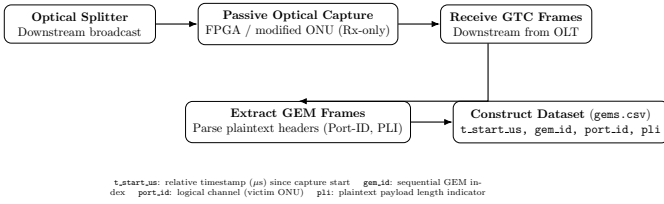


Fig. 2. Attacker-side observation pipeline: passive optical capture under the same splitter enables reception of downstream GTC frames, extraction of GEM frames, and construction of a PLI-based dataset (`gems.csv`).

IV. TRACE GENERATION (125)

To demonstrate the threat model in our experimental environment, We developed software based GPON downstream trace generator. From a PCAP of encrypted video traffic, we (i) map Ethernet frames into GEM payloads (fragmenting to satisfy the 12-bit PLI limit), (ii) generate plaintext GEM headers (PLI/Port-ID/PTI/HEC), (iii) encrypt payload bytes with AES-CTR (header remains plaintext), and (iv) pack the GEM stream into fixed-length 125 downstream frames consistent with the configured line rate. The attacker signal is then either the GEM-level PLI sequence or the per-frame occupancy series. Large Ethernet packets may generate multiple consecutive GEM fragments; the PLI values in our traces therefore reflect per-fragment payload lengths (the attacker-visible observable).

V. EXPERIMENT AND PRELIMINARY RESULTS

First, we played each of 10 videos (three runs each) on Ubuntu 24.04 under a fixed playback quality and resolution setting(720p). To minimize interference, we restricted other background traffic (e.g., VNC, SSH) while playing video. For each run, we captured packet data (PCAP) for 180 seconds. Then, we replayed each PCAP file in our simulator to record the resulting PLI values and timestamps as CSV traces. We preserve original PCAP packet timestamps to retain idle periods and burstiness; each synthesized GEM is assigned `t_start_us` derived from the PCAP time base (μ s resolution). For analyses that require alignment, we additionally aggregate GEMs into 125- μ s bins matching the GPON downstream

frame period. These traces approximate the payload length observations available to a real same-splitter adversary and enable systematic evaluation of PLI based inference.

After producing the dataset, we visualized cumulative PLI–Time Plot based on our dataset(Publicly Available on our github repository) [3]. We observed a clear correlation between each video and its cumulative PLI over time graph; traces from the same video show highly consistent slopes across runs. We use run1-2 as a reference database and treat run3 as a query trace (run3-only inter comparison).

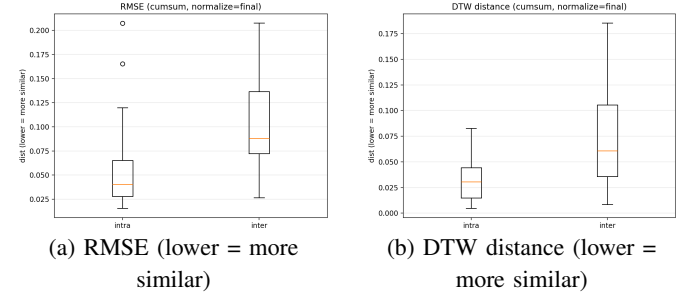


Fig. 4. Distance distributions between same-video (intra) and different-video (inter) cumulative PLI traces. Same-video pairs tend to yield smaller distances, with partial overlap due to ABR variability and buffering.

We further quantify the similarity between traces using distance-based metrics on the normalized cumulative PLI time series. Same-video pairs yield smaller distances than different-video pairs Same-video pairs yield smaller distances than different-video pairs (median RMSE: 0.040 vs. 0.088, AUC=0.813; median DTW: 0.031 vs. 0.061, AUC=0.776).

VI. CONCLUSION AND DISCUSSION

Despite GEM Payload Encryption, GPON Network is still vulnerable to inferring streaming activity because PLI in GEM headers can be exploited to infer a victim's video activity(e.g., Youtube). Potential mitigations include traffic shaping and obfuscating GEM metadata. All code and experimental dataset is publicly available on our github repository. [3]

Future Work Future Work will be conducted in testbed with Real OLT, ONU and Optical Splitter. We will use FPGA device to capture GEM frames to parse downstream frames. We will expand to more videos and longer captures(while changing various resolution, bitrate); report top-1 identification accuracy and confusion matrices (not only pairwise AUC).

Extent to XGS-PON Networks. We assume that XGS-PON are subject to analogous privacy risks because XGEM Frame that XGS-PON uses has Payload Length Indicator Field like GEM. however, whether the same side channel persists depends on which XGEM header fields remain plaintext and how OLT scheduling differs. We will validate this by repeating our capture-and-matching evaluation on an XGS-PON testbed.

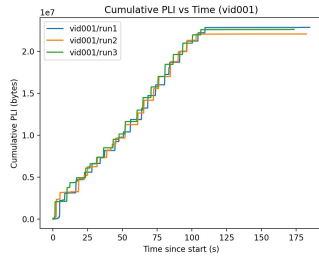


Fig. 3. Cumulative PLI over time for one video across three runs (run1–run3). Traces show closely matching slopes, indicating high within-video consistency under fixed settings. Other traces are available on our GitHub repository [3].

REFERENCES

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